

Traffic Signal Recognition

Weekly Report #1 | University of Houston – Clear Lake

Course	CENG 4266 — Senior Project
Project	Traffic Signal Recognition
Report	Weekly Report #1
Period	August 18 – August 24, 2026
Prepared	August 25, 2026
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Progress This Week

- Drafted the project proposal describing our goal of building a self-contained board that recognizes traffic signs in real time.
- Wrote up the hardware plan: choose a small compute module and design a custom carrier board around it, handling power, connectors, camera, and I/O ourselves.
- Noted that we will use the module's reference design as a starting point so the carrier board is not designed blind.
- Described the software plan: train our own vision model from scratch on a public traffic-sign dataset and run it directly on the board with no internet or cloud.
- Defined how we will measure success — recognition accuracy, per-image processing speed on the board, and power draw.
- Laid out the two-semester timeline in the proposal: hardware selection, board design, and model training in the fall; integration, bring-up, and testing in the spring.
- Pending: Send the initial proposal to Professor Nguyen for feedback.